Classification Systems in Design

Even with many resources available for site investigation, there still can remain problems in applying theories in practical engineering circumstances. Considering the three main design approaches for engineering rock mechanics – analytical, observational and empirical, rock mass classifications today form an integral part of the most predominant design approach, the empirical design method.

Indeed, on many underground construction, tunnelling and mining projects, rock mass classifications have provided the only systematic design aid in an otherwise haphazard "trial-and-error" procedure.

Rock Mass Classification

The objectives of rock mass classifications are to:

- Identify the most important parameters influencing the rock mass.
- Divide a rock mass formation into groups of similar behaviour.
- Provide a basis for understanding the characteristics of each rock mass class.
- Relate experiences of rock conditions at one site to those at another.
- Derive quantitative data and guidelines for engineering design.
- Provide a common basis for communication between geologists and engineers.

The boundaries of the structural regions usually coincide with a major structural feature such as a fault or with a change in rock type. In some cases, significant changes in discontinuity spacing or characteristics, within the same rock type, may necessitate the division of the rock mass into a number of small structural regions.

Rock Mass Classification

These objectives suggest the three main benefits of rock mass classifications:

- Improving the quality of site investigations by calling for the minimum input data as classification parameters.
- Providing quantitative information for design purposes.
- Enabling better engineering judgment and more effective communication on a project.

Engineering properties of large mass

Rock Mass Quality

Engineering classification can be used for assess rock mass quality for estimating overall engineering rock mass behviour.

RQD can also used as a measure for rock mass quality:

The RQD is a modified core recovery percentage in which all the pieces of sound core over 4-inches (100-mm) long are summed and divided by the length of the core run.

The RQD is an index of rock quality in that problematic rock that is highly weathered, soft, fractured, sheared, and jointed is counted against the rock mass.

Engineering properties of large mass

Rock Mass Rating

For the analysis of failure through the rock mass it is necessary to determine the friction angle and cohesion of the rock mass itself. However, testing of representative rock mass samples is difficult because of sample disturbance and equipment size limitations. Thus the preferred method has been to derive empirical values of friction and cohesion based on rock mass rating systems that have been calibrated from experience.

Engineering properties of large mass

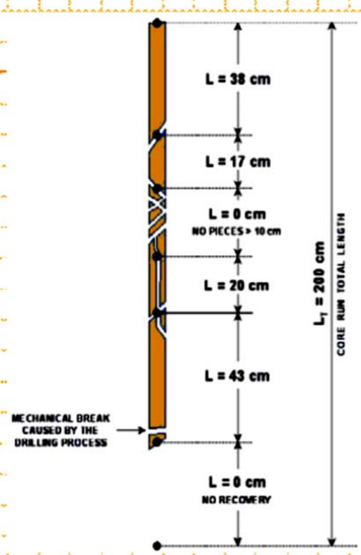
Rock Mass Rating

Some of them were extended for the use in rock engineering, and those that are commonly used include

- Tunnelling Quality Index (Q) (Barton)
- Rock Mass Rating (RMR) (Bieniawski)
- Mining Rock Mass Rating (MRMR) (Laubscher)
- Geological Strength Index (GSI) (Hoek)
- Rock Mass Index (RMi) (Palmström)

Name of classification	Originator and date	Country of origin	Applications
Rock loads	Terzaghi, 1946	USA	Tunnels with steel supports
Stand-up time	Lauffer, 1958	Austria	Tunnelling
Rock quality designation (RQD)	Deere, 1964	USA	Core logging, Tunnelling
Rock Structure Rating (RSR)	Wickham et al. 1972	USA	Tunnelling
Geomechanical classification (RMR) system	Bieniawski, 1973	S. Africa & USA	Tunnels, mines, foundations
Q-system	Barton, et al. 1974	Norway	Tunnelling, large chambers
CMRS-ISM Geomechanical classification(RMR)	CMRI, 1987	India	Mine workings
Mining Rock Mass Rating (MRMR)	Laubscher. 1977	S. Africa	Mining Excavations
Rock Mass Index (RMi)	Palmström. 1995 ,2000	Norway	Tunnelling, Underground Excavations
Coal Mine Roof Rating (CMRR)	(Molinda & Mark) ,1993 US Bureau of Mines	USA	Coal mine working
Geological Strength Index(GSI)	Hoek,1994	USA	Tunnelling, Underground Excavations
Coal Measure Classification(CMC)	Whttiles,2006	UK	Coal mine working

Rock Quality Designation (RQD)



$$RQD = \frac{\sum Rock\ Pieces \geq 10\ cm}{Core\ Run\ Total\ Length} \times 100\,(\%)$$

$$RQD = \frac{38 + 17 + 0 + 20 + 43 + 0}{200} \times 100\,(\%)$$

$$RQD = 59\,\%\ \ (FAIR)$$

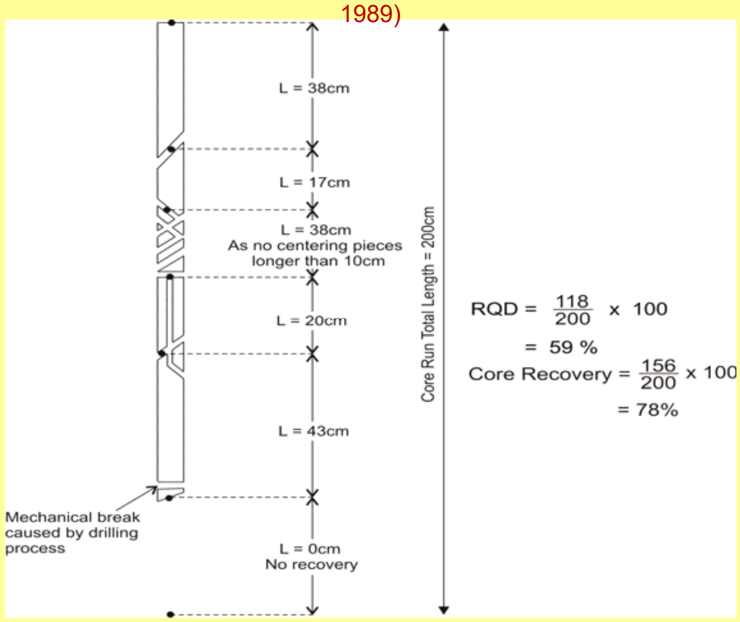
RQD (%)	Geotechnical Quality
< 25	VERY POOR
25 to 50	POOR
50 to 75	FAIR
75 to 90	GOOD
90 to 100	EXCELLENT

Correlation between RQD and rock quality

S. No.	RQD (%)	Rock Quality
1	<25	Very poor
2	25-50	Poor
3	50-75	Fair
4	75-90	Good
5	90-100	Excellent

RQD is perhaps the most commonly used method for characterizing the degree of jointing in borehole cores, although this parameter also may implicitly include other rock mass features like weathering and 'core loss' (Bieniawski, 1989).

Procedure for measurement and calculation of rock quality designation RQD (Deere, 1989)



- The geomechanics classification or the rock mass rating (RMR) system was initially developed at the South African Council of Scientific and Industrial Research (CSIR) by Bieniawski (1973) on the basis of his experiences in shallow tunnels in sedimentary rocks.

- To apply the geomechanics classification system, a given site should be divided into a number of geological structural units in such a way that each type of rock mass is represented by a separate geological structural unit.

ESTIMATION OF ROCK MASS RATING (RMR)

The following six parameters (representing causative factors) are determined for each of the structural unit:

1) Uniaxial compressive strength of intact rock material,
2) Rock quality designation RQD,
3) Joint or discontinuity spacing,
4) Joint condition,
5) Ground water condition, and
6) Joint orientation.

- When mixed quality rock conditions are encountered at the excavated rock face, such as 'good quality' and 'poor quality' being present in one exposed area, it is essential to identify the 'most critical condition' for the assessment of the rock strata.

ESTIMATION OF ROCK MASS RATING (RMR)

- On the basis of RMR values for a given engineering structure, the rock mass is classified in five classes named as
  - very good (RMR 100-81),
  - good (80-61),
  - fair (60-41),
  - poor (40-21) and
  - very poor (<20)

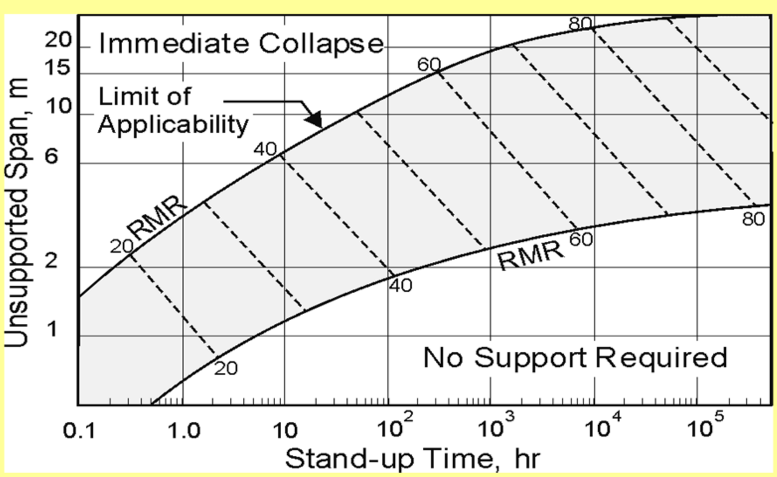
Table 10: Design parameters & engineering properties of rock mass (Bieniawski, 1993)

Sl. No.	Parameter/ Properties of Rock Mass	Rock Mass Rating (Rock Class)				
		100-81 (I)	80-61 (II)	60-41 (III)	40-21 (IV)	<20 (V)
1.	Classification of rock mass	Very good	Good	Fair	Poor	Very poor
2.	Average stand-up time	20 years for 15 m span	1 year for 10m span	1 week for 5 m span	10 hrs for 2.5m span	30 min. for 1 m span
3.	Cohesion of rock mass (MPa)*	> 0.4	0.3-0.4	0.2-0.3	0.1-0.2	< 0.1
4.	Angle of internal friction of rock mass	> 45°	35°-45°	25°-35°	15°-25°	< 15°
5.	Allowable bearing pressure (T/m^2)	600-440	440-280	280-135	135-45	45-30
6.	Safe cut slope (°) (Waltham, 2002)	> 70	65	55	45	< 40

** These values are applicable to slopes only in saturated and weathered rock mass*

Note: During earthquake loading, the above values of allowable bearing pressure may be increased by 50 percent in view of rheological behaviour of rock masses.

Stand-up time vs. unsupported span for various rock mass classes according to RMR (Bieniawski, 1984)



Rock Mass Classification: RMR

The Rock Mass Rating (RMR) system was developed in 1973 in South Africa by Prof. Z.T. Bieniawski. The advantage of his system was that only a few basic parameters relating to the geometry and mechanical conditions of the rock mass were required.

In applying this system, the rock mass is divided into a number of structural domains and each is classified separately. Because parameters are not equally important, weighted ratings are allocated.

A. CLASSIFICATION PARAMETERS AND THEIR RATINGS

	Parameter		Ranges of values						
1	Strength of intact rock material	Point-load strength index (MPa)	>10	4 - 10	2 - 4	1 - 2	For this low range, uniaxial compressive test is preferred		
		Uniaxial compressive strength (MPa)	>250	100 - 250	50 - 100	25 - 50	5 - 25	1 - 5	<1
		Rating	15	12	7	4	2	1	0
2	Drill core quality RQD (%)		90 - 100	75 - 90	50 - 75	25 - 50	<25		
		Rating	20	17	13	8	3		
3	Spacing of discontinuities		>2m	0.6 - 2m	200 - 600mm	60 - 200mm	<60mm		
		Rating	20	15	10	8	5		
4	Condition of discontinuities		Very rough surfaces Not continuous No separation Unweathered wall rock	Slightly rough surfaces Separation <1mm Slightly weathered wall rock	Slightly rough surfaces Separation <1mm Highly weathered wall rock	Slickensided surfaces or Gouge <5mm thick or Separation 1 - 5mm Continuous	Soft gouge >5mm thick or Separation >5mm Continuous		
		Rating	30	25	20	10	0		
5	Groundwater	Inflow per 10m tunnel length (l/min)	None	<10	10 - 25	25 - 125	>125		
		ratio (joint water pressure)/(major principal stress)	0	<0.1	0.1 - 0.2	0.2 - 0.5	>0.5		
		General conditions	Completely dry	Damp	Wet	Dripping	Flowing		
		Rating	15	10	7	4	0		

B. RATING ADJUSTMENT FOR DISCONTINUITY ORIENTATIONS

Strike and dip orientations		Very favourable	Favourable	Fair	Unfavourable	Very Unfavourable
Ratings	Tunnels & mines	0	-2	-5	-10	-12
	Foundations	0	-2	-7	-15	-25
	Slopes	0	-5	-25	-50	

Bieniawski (1989)

Rock Mass Classification: Q-System

The Q-system of rock mass classification was developed in 1974 in Norway by Prof. N. Barton. The system was proposed on the basis of an analysis of 212 tunnel case histories from Scandinavia.

$$Q = \frac{\mathrm{RQD}}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{\mathrm{SRF}}$$

where

- RQD = rock quality designation,
- J_n = joint set number (related to the number of discontinuity sets),
- J_r = joint roughness number (related to the roughness of the discontinuity surfaces),
- J_a = joint alteration number (related to the degree of alteration or weathering of the discontinuity surfaces),
- J_w = joint water reduction number (relates to pressures and inflow rates of water within the discontinuities), and
- SRF = stress reduction factor (related to the presence of shear zones, stress concentrations and squeezing and swelling rocks).

… the motivation of presenting the Q-value in this form is to provide some method of interpretation for the 3 constituent quotients.

Q-SYSTEM

$$Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$$

where,

RQD = Deere's Rock Quality Designation ≥ 10,

J_n = Joint set number,

J_r = Joint roughness number for critically oriented joint set,

J_a = Joint alteration number for critically oriented joint set,

J_w = Joint water reduction factor,

SRF = Stress reduction factor to consider in situ stresses and according to the observed tunnelling conditions, and

The goal of Q-system is to characterise the rock mass and preliminary empirical design of support system for tunnels and caverns.

$$\frac{RQD}{J_n} = \text{Degree of jointing (or block size)}$$

$$\frac{J_r}{J_a} = \text{Joint friction (inter-block shear strength)}$$

$$\frac{J_w}{SRF} = \text{Active stress}$$

Rock Mass Classification: Q-System

The first quotient is related to the rock mass geometry. Since RQD generally increases with decreasing number of discontinuity sets, the numerator and denominator of the quotient mutually reinforce one another.

$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

The second quotient relates to "inter-block shear strength" with high values representing better 'mechanical quality' of the rock mass.

$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

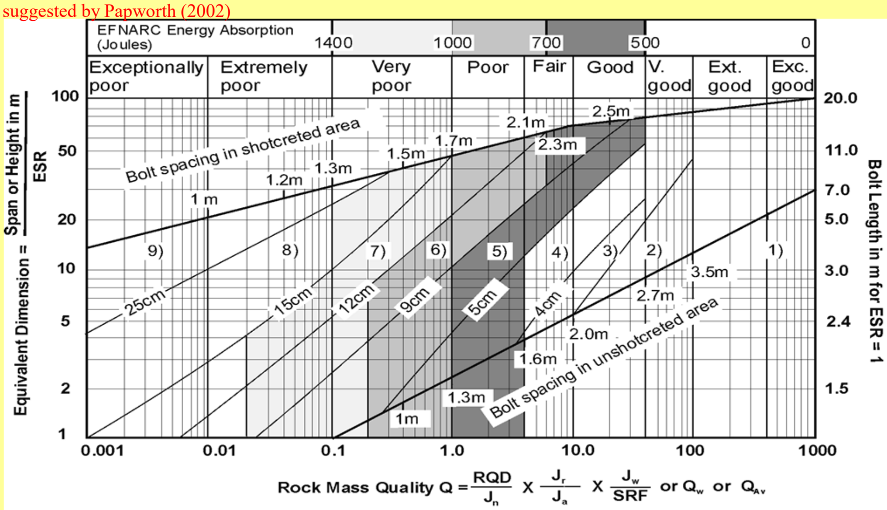
The third quotient is an 'environment factor' incorporating water pressures and flows, the presence of shear zones, squeezing and swelling rock and the *in situ* stress state. The quotient increases with decreasing water pressure and favourable *in situ* stress ratios.

$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

Classification of rock mass based on Q - values

Q	Group	Classification
0.001 - 0.01 0.01 - 0.1	3	Exceptionally poor Extremely poor
0.1 - 1 1 - 4 4 - 10	2	Very poor Poor Fair
10 - 40 40 - 100 100 - 400 400 - 1000	1	Good Very good Extremely good Exceptionally good

Grimstad and Barton (1993) chart for the design of support including the required energy absorption capacity of SFRS suggested by Papworth (2002)



REINFORCEMENT CATEGORIES

1) Unsupported
2) Spot bolting, sb
3) Systematic bolting, B
4) Systematic bolting (and unreinforced shotcrete, 4 to10cm, B(+S)
5) Fiber reinforced shotcrete and bolting, 5 to 9cm, S(fr)+B
6) Fiber reinforced shotcrete and bolting, 9 to 12cm, S(fr)+B
7) Fiber reinforced shotcrete and bolting, 12 to 15cm, S(fr)+B
8) Fiber reinforced shotcrete > 15cm, reinforced ribs of shotcrete and bolting, S(fr), RRS+B
9) Cast concrete lining,CCA

APPLICATIONS OF RMR & Q-SYSTEM

- Average stand-up time for arched roof
- Cohesion and angle of internal friction
- Modulus of deformation
- Estimation of support pressure

Rock Mass Classification – Examples

- ✓ massive, strong rock
- ✓ low stress regime
- ✓ note lack of ground support
- ✓ RMR = 90 (very good rock)
- ✓ Q = 180 (extremely good rock)

Courtesy - Golder Associates

Rock Mass Classification – Examples

- ✓ blocky rock
- ✓ low stress regime
- ✓ minimal but systematic ground support
- ✓ RMR = 70 (good rock)
- ✓ Q = 15 (good rock)

Courtesy - Golder Associates

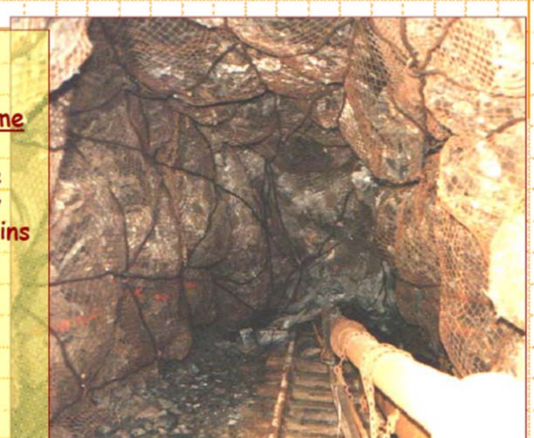
Rock Mass Classification – Examples

- ✓ weak/foliated rock
- ✓ low stress regime
- ✓ note lack of ground support
- ✓ RMR = 40 (poor to fair rock)
- ✓ Q = 0.9 (v.poor to poor rock)

Courtesy - Golder Associates

Rock Mass Classification – Examples

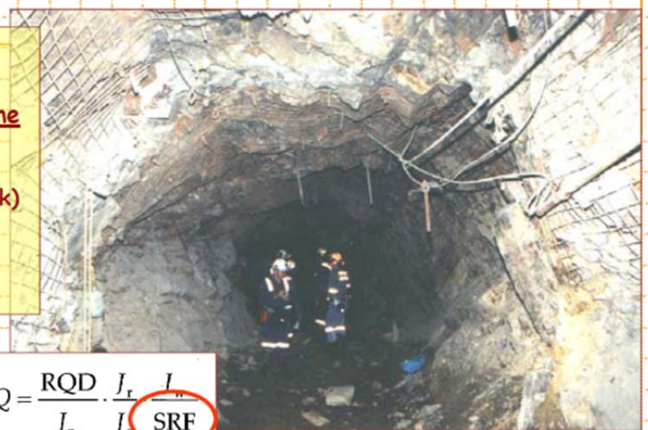
- ✓ massive, strong rock
- ✓ extremely high stress regime
- ✓ rockburst failure, complete closure of drift, extremely heavy support, screen retains failed rock
- ✓ RMR = 80 (good to v.good rock)
- ✓ Q = 0.5 (very poor rock)



Courtesy - Golder Associates

Rock Mass Classification – Examples

- blocky rock
- high stress regime
- RMR = 40 (poor to fair rock)
- Q = 0.8 (very poor rock)

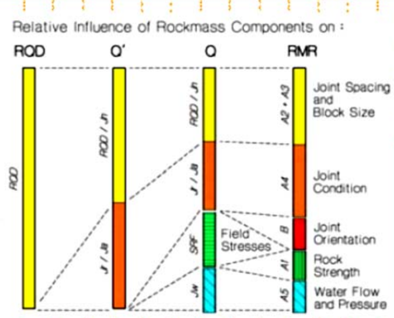


$$Q = \frac{RQD}{J_n} \cdot \frac{J_r}{J_a} \cdot \frac{J_w}{SRF}$$

Courtesy - Golder Associates

Rock Mass Classification – RMR versus Q

Hutchinson & Diederichs (1996)

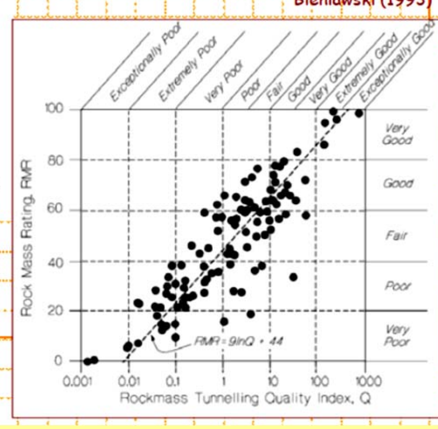


$$RMR = 9\ ln_e Q + 44$$

$$RMR = 21\ log_{10} Q + 44$$

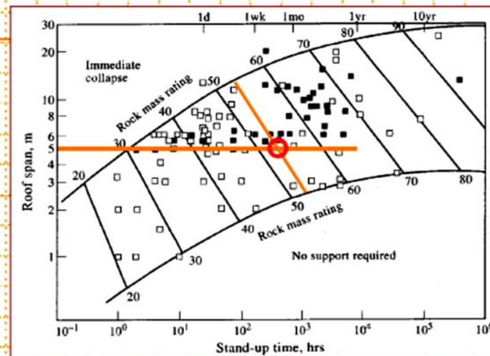
$$Q = 10^{(RMR - 44)/21}$$

Bieniawski (1993)



Application of Classification Systems

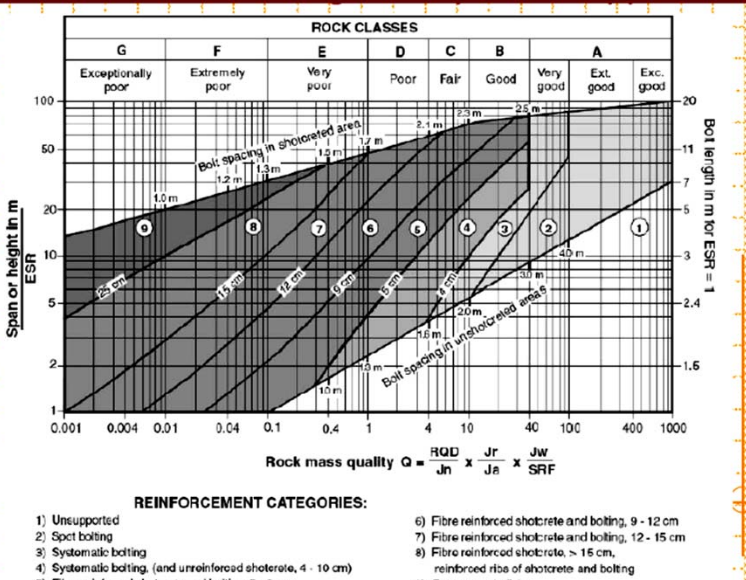
Rock masses can be extremely complex, making the derivation of predictive equations difficult. As an alternative, rock mass classification methods have been calibrated against large databases of case histories to provide guidelines for support design.



Bieniawski (1989)

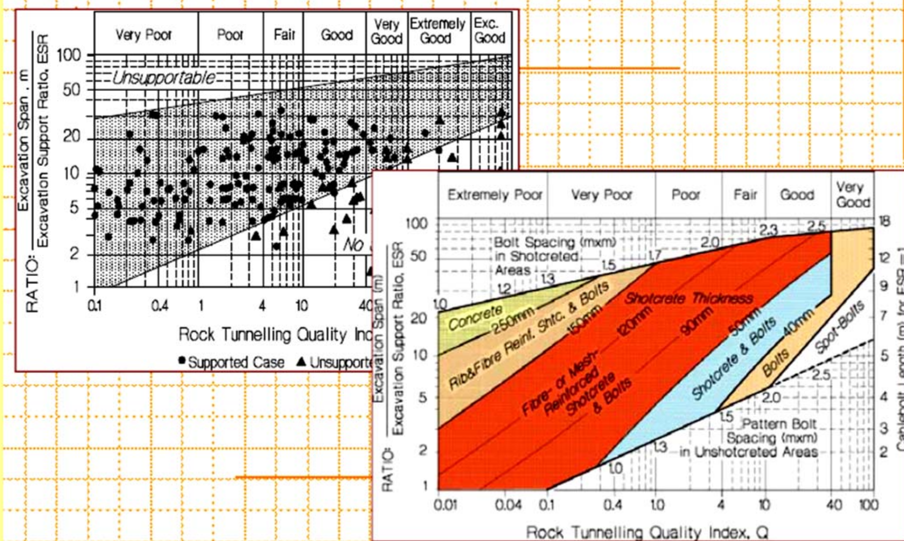
Empirical design of stand-up time, the duration within which an unsupported excavation will remain serviceable, after which significant caving and failure may occur. The database used in its development examined 351 civil tunnel and underground mine case histories.

Experience-Based Design: Empirical Approaches



Grimstad & Barton (1993)

Experience-Based Design: Empirical Approaches



Kaiser *et al.* (2000)

GROUNDWATER

The presence of groundwater in underground operations often creates serious problems. The most important is generally a reduction in stability of the excavation.

The amount of groundwater present, the rate at which it will flow through the rock, the effect it may have on stability depends on many factors.

204

GROUNDWATER

The most important of these are

- topography of the area
- precipitation
- temperature variation
- permeability of the rock mass and fragmentation and
- orientation of structural discontinuities in the rock

205

GROUNDWATER

Groundwater may affect the mechanical performance of a rock mass in two ways.

a)Pore pressure: Water under pressure in the joints defining rock blocks reduces the normal effective stress between the rock surfaces, and the effect of fissure or pore water under pressure is to reduce the ultimate strength of the mass, when compared with the drained condition.

GROUNDWATER

b)Presence of minerals: A more subtle effect of groundwater on rock mechanical properties may arise from the deleterious action of water on particular rocks and minerals. For example, rock may contain clay may soften in the presence of groundwater, reducing the strength and increasing the deformability of the rock mass. Argillaceous rocks, such as shales and argillitic sandstones, also demonstrate marked reductions in material strength following infusion with water.

GROUNDWATER

(c) WEATHERING OF ROCK: Rocks may be broken down by the combined effects of temperature, air, water and associated chemical activity.

Change in water content of some rocks, particularly shale, can cause accelerated weathering with a resulting decrease stability.

Chemical weathering is a generally slower process, but has the potential to significantly affect rock mass strength. It often has no effect during the working life of the open pit, but may be a factor affecting the long term stability of slopes on abandonment. Weathered rock may commonly be present prior to excavation.

208

GROUNDWATER

(d) FREEZING OF GROUNDWATER BLOCK DRAINAGE

The greatest reduction in stability can be expected to occur during and shortly after spring snow melt, following heavy rainfall, and during the late stages of winter (locations where temperature causes freezing of water).

In the snow fall area, it has frequently been assumed that winter conditions should be conducive to stability since the surface of the rock is frozen.

209

GROUNDWATER

However, the water which normally is free to flow out through the cracks in the rock during the summer period becomes frozen during the winter and greatly reduces the permeability of the surface of the rock. This can result in a build-up of very high pore water pressures behind the face. As a result, the shear strength of the rock diminishes rapidly and sudden failure of the rock may take place.

210

GROUNDWATER

(e) EROSION OF SOIL & INFILLING MATERIAL:

Flow of water can give rise to erosion infilling material in filled jointed rock at surface and in subsurface which may cause reduction in stability of opening.

Measurement of water pressure

For a reliable estimate of stability of opening in rock or to control drainage, it is essential that water pressures within the rockmass should be measured.

A **piezometer** is a small-diameter observation well used to measure the hydraulic head (water pressure) of groundwater in soil or rock stratum.

A **piezometer** is a device used to determine static water pressure in a system by measuring the height to which a column of the liquid rises against gravity, or a device which measures the pressure (more precisely, the piezometric head) of groundwater at a specific point.

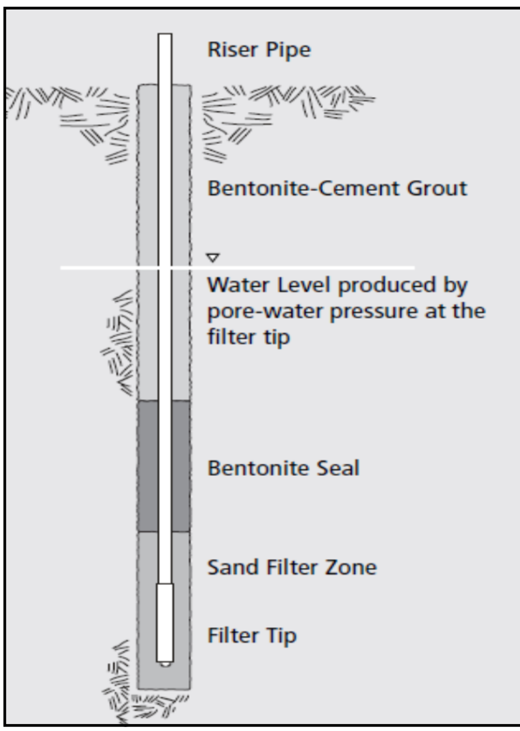
Such measurments are most conveniently carried out by **piezometer** installed in boreholes.

212

Measurement of water pressure

The first piezometers in geotechnical engineering were open wells or standpipes (sometimes called **Casagrande piezometers**) installed into an aquifer. A Casagrande piezometer will typically have a solid casing down to the depth of interest, and a slotted or screened casing within the zone where water pressure is being measured. The casing is sealed into the drillhole with clay, bentonite or concrete to prevent groundwater leaking to the surface.

213



Water level in standpipe (Casagrande) piezometer is produced by pore-water pressure at the filter tip.

214

Measurement of water pressure

Piezometers in durable casings can be buried or pushed into the ground to measure the groundwater pressure at the point of installation. The pressure gauges (transducer) can be vibrating-wire, pneumatic, or strain-gauge in operation, converting pressure into an electrical signal. These piezometers are cabled to the surface where they can be read by data loggers or portable readout units, allowing faster or more frequent reading than is possible with open standpipe piezometers.

215

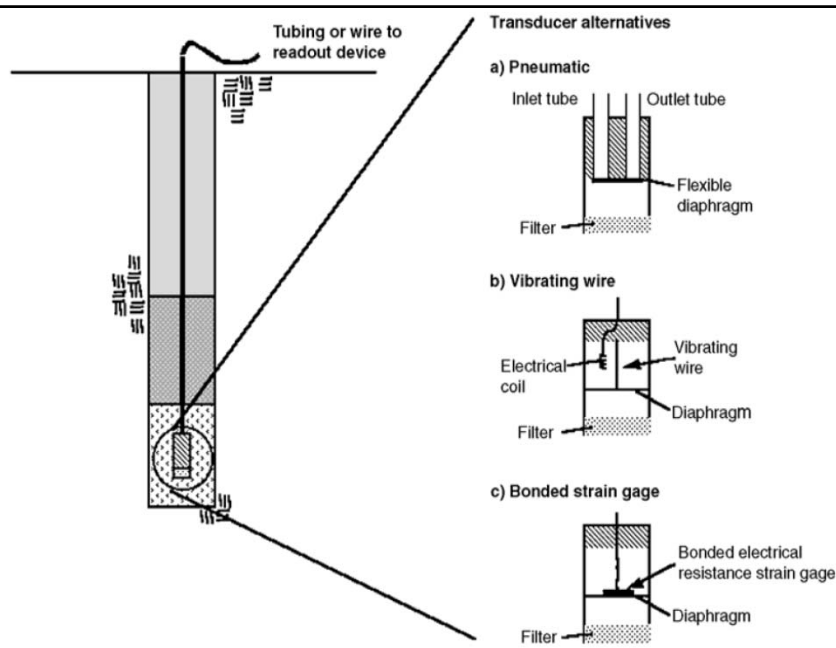


FIGURE 28.8 Schematic of (a) pneumatic, (b) vibrating wire piezometer, and (c) bonded strain gage electrical resistance.

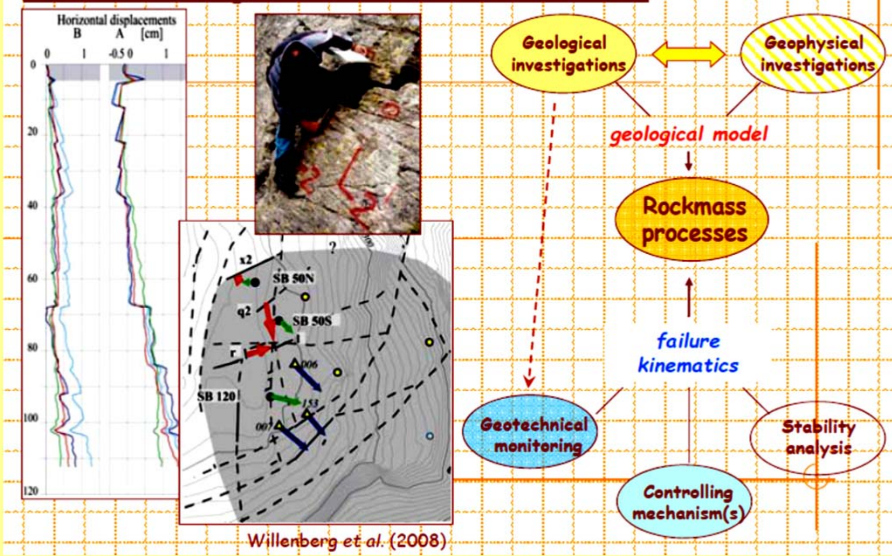
TABLE 28.2 Advantages and Disadvantages of Different Types of Piezometers

Piezometer	Advantages	Disadvantages
Open standpipe	• Highly reliable • Low cost • Simple to install • Easy to read • Can perform permeability tests • Can obtain groundwater samples	• Long time lag especially in low permeability soils • Interference with construction when used in embankments on soft soils • Susceptible to damage by construction equipment
Hydraulic twin tubing	• Reliable for long term monitoring • Flushing possible to remove air • Can perform permeability tests • Can measure negative pore pressure	• Complex installation • Very difficult and costly to implement automated data acquisition • Periodic flushing required • Limitation on elevation above piezometric head at which measurement is made • Freezing problems

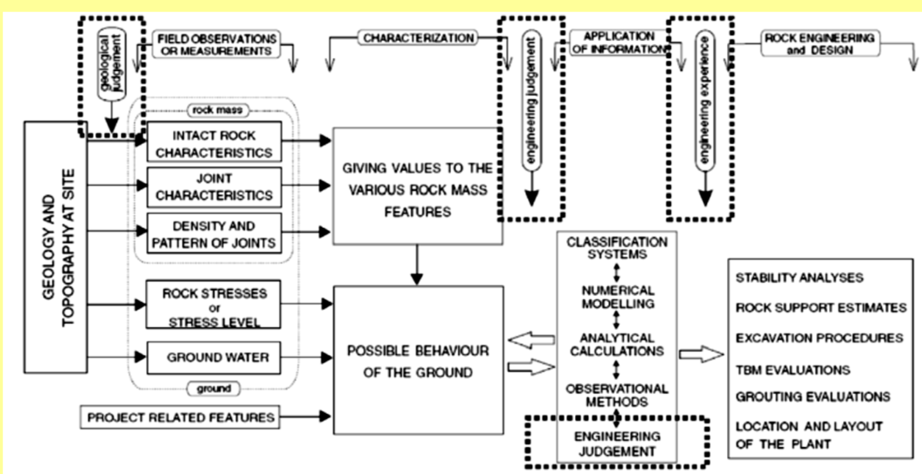
TABLE 28.2 Advantages and Disadvantages of Different Types of Piezometers

Piezometer	Advantages	Disadvantages
Pneumatic	• Short time lag (provided tubing not excessively long) • No freezing problems • Can be used in corrosive environments	• Very difficult and costly to implement automated data acquisition • Reading time increases with length of tubing
Electrical resistance	• Simple to install • Short time lag • Simple to read • Easy to implement automated data acquisition • Some types can be used for dynamic measurements • Can measure negative pore pressure • No freezing problems	• Moderate to high cost • Susceptible to damage by lightning • Susceptible to damage by moisture • Low electrical output
Vibrating wire	• Simple to install • Short time lag • Very accurate • Simple to read • Easy to implement automated data acquisition • Can measure negative pore pressure • No freezing problems	• Moderate to high cost • Susceptible to damage by lightning

Site Investigation & Data Collection



Willenberg *et al.* (2008)



Improvements in site characterisation methodology and techniques, and in the interpretation of the data are of primary research requirements for all forms of rock engineering.